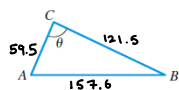


5.6
1) Use Law of cosines to find θ

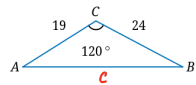


$$\cos \theta = \frac{59.5^2 + 121.5^2 - 157.6^2}{2(59.5)(121.5)}$$

$$\theta = \cos^{-1} \left[\frac{59.5^2 + 121.5^2 - 157.6^2}{2(59.5)(121.5)} \right]$$

$$\theta \approx 116.9^\circ$$

2) Solve $\triangle ABC$



$$c = \sqrt{19^2 + 24^2 - 2(19)(24)\cos(120^\circ)}$$

$$c \approx 37.3 \text{ units}$$

$$\frac{\sin A}{24} = \frac{\sin(120^\circ)}{c}$$

$$A = \sin^{-1} \left[\frac{24 \sin(120^\circ)}{c} \right]$$

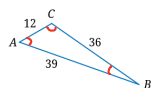
$$\angle A \approx 33.8^\circ$$

$$\frac{\sin B}{19} = \frac{\sin(120^\circ)}{c}$$

$$B = \sin^{-1} \left[\frac{19 \sin(120^\circ)}{c} \right]$$

$$\angle B \approx 26.2^\circ$$

3) Solve $\triangle ABC$



$$\angle A = \cos^{-1} \left[\frac{12^2 + 39^2 - 36^2}{2 \cdot 12 \cdot 39} \right]$$

$$\angle A \approx 66.8^\circ$$

$$\angle B = \cos^{-1} \left[\frac{39^2 + 36^2 - 12^2}{2 \cdot 39 \cdot 36} \right]$$

$$\angle B \approx 17.8^\circ$$

$$\angle C = \cos^{-1} \left[\frac{12^2 + 36^2 - 39^2}{2 \cdot 12 \cdot 36} \right]$$

$$\angle C \approx 95.4^\circ$$

4) Solve $\triangle ABC$

$$b = 124, c = 163, \angle B = 46^\circ$$

$$\text{SSA, } \angle B < 90^\circ$$

$$c \sin B \approx 117$$

$$c > b > c \sin B \rightarrow 2 \text{ solutions}$$

$$\frac{124}{\sin(46^\circ)} = \frac{163}{\sin(C_1)}$$

$$C_1 = \sin^{-1} \left[\frac{163 \sin(46^\circ)}{124} \right]$$

$$\angle C_1 \approx 71.0^\circ$$

$$A_1 = 180^\circ - 46^\circ - C_1$$

$$\angle A_1 \approx 63.0^\circ$$

$$\frac{124}{\sin(46^\circ)} = \frac{a_1}{\sin(A_1)}$$

$$a_1 = \frac{124 \sin(A_1)}{\sin(46^\circ)}$$

$$a_1 \approx 153.6 \text{ units}$$

$$C_2 = 180^\circ - C_1$$

$$\angle C_2 \approx 109.0^\circ$$

$$A_2 = 180^\circ - 46^\circ - C_2$$

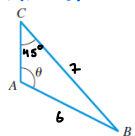
$$\angle A_2 \approx 25.0^\circ$$

$$\frac{a_2}{\sin(A_2)} = \frac{124}{\sin(46^\circ)}$$

$$a_2 = \left[\frac{124 \sin(A_2)}{\sin(46^\circ)} \right]$$

$$a_2 \approx 72.9 \text{ units}$$

- 5) Find θ , assume $\theta > 90^\circ$.

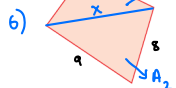


$$\frac{\sin \theta}{7} = \frac{\sin(45^\circ)}{6}$$

$$\theta = \sin^{-1} \left[\frac{7 \sin(45^\circ)}{6} \right]$$

$$\theta \approx 55.6$$

$$180^\circ - \theta \approx 124.4^\circ$$



$$x = \sqrt{6^2 + 7^2 - 2(6)(7) \cos(100^\circ)}$$

$$s_1 = \frac{1}{2}(6+7+x) = y$$

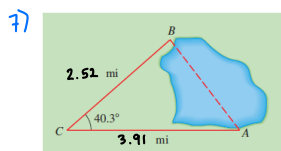
$$s_2 = \frac{1}{2}(8+9+x) = z$$

$$A_1 = \sqrt{y(y-6)(y-7)(y-x)}$$

$$A_2 = \sqrt{z(z-8)(z-9)(z-x)}$$

$$A = A_1 + A_2$$

$$A \approx 54.84 \text{ in}^2$$

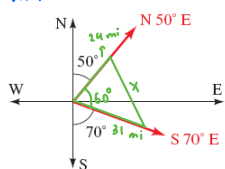


Find distance in mi from A to B

$$AB = \sqrt{2.52^2 + 3.91^2 - 2(2.52)(3.91) \cos(40.3^\circ)}$$

$$\approx 2.57 \text{ mi}$$

- 6) Two boats leave same port at same time. One travels at speed of 24 mi/h in direction $N 50^\circ E$ and other travels at speed of 31 mi/h in direction $S 70^\circ E$.

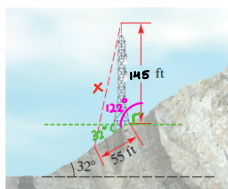


$$x = \sqrt{24^2 + 31^2 - 2(24)(31) \cos(60^\circ)}$$

$$x \approx 28 \text{ mi}$$

How far apart in mi are the 2 boats after 1 hour?

- 7) 145-ft tower located on side of mountain that is inclined 32° to the horizontal. guy wire is to be attached top of tower and anchored at point 55 ft downhill from base of tower.

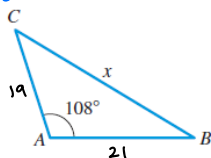


Find shortest length of wire needed.

$$x = \sqrt{145^2 + 55^2 - 2(145)(55) \cos(122^\circ)}$$

$$x \approx 180 \text{ ft}$$

- 10) Use law of cosines to find x.



$$x = \sqrt{19^2 + 21^2 - 2(19)(21) \cos(108^\circ)}$$

$$x \approx 32.4 \text{ in}$$

- 11) Solve $\triangle ABC$

$$b = 68, c = 34, \angle A = 72^\circ$$

SAS

$$a = \sqrt{68^2 + 34^2 - 2(68)(34) \cos(72^\circ)}$$

$$a \approx 66.0 \text{ in}$$

$$\frac{a}{\sin(72^\circ)} = \frac{68}{\sin B}$$

$$B = \sin^{-1} \left[\frac{68 \sin(72^\circ)}{a} \right]$$

$$\angle B \approx 78.6^\circ$$

$$\angle C = 180^\circ - 72^\circ - \angle B$$

$$\angle C \approx 29.4^\circ$$

12) Solve $\triangle ABC$

$$a=23, b=46, c=24$$

SSS

$$A = \cos^{-1} \left[\frac{46^2 + 24^2 - 23^2}{2(46)(24)} \right]$$

$$\angle A \approx 11.6^\circ$$

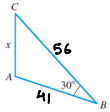
$$B = \cos^{-1} \left[\frac{23^2 + 24^2 - 46^2}{2(23)(24)} \right]$$

$$\angle B \approx 156.3^\circ$$

$$C = 180^\circ - \angle A - \angle B$$

$$\angle C \approx 12.1^\circ$$

13) Find x



$$x = \sqrt{41^2 + 56^2 - 2(41)(56) \cos(30^\circ)}$$

$$x \approx 29.0 \text{ un}$$

14) Find area A of $\triangle$ whose sides have given lengths

$$a=11, b=103, c=104$$

$$s = \frac{1}{2} (11 + 103 + 104) = 109$$

$$A = \sqrt{109(109-11)(109-103)(109-104)}$$

$$A \approx 566 \text{ un}^2$$

15) $\triangle$ whose sides have given lengths

$$a=25, b=15, c=20$$

Using Heron's formula find semiperimeter s of the $\triangle$

$$s = \frac{1}{2} (25 + 15 + 20) = 30 \text{ un}$$

find area of $\triangle$

$$A = \sqrt{30(30-25)(30-15)(30-20)}$$

$$A = 150 \text{ un}^2$$